

7 Troubleshooting

Troubleshooting using error codes

7.4 Troubleshooting using error codes

7.4.1 List of error codes

Software: 7528678

Error codes

No.	Description	Felorsak(er)	Action
2:002	Parameters set to default values. At start-up, it was detected that one or more parameters were not within acceptable intervals; they have therefore been reset to default values Truck behaviour Not influenced	- New software has been loaded in the truck, adding a new parameter, or it has changed the parameter limit values - Corrupt memory A5	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 2) Replace A5 See section <i>Removing the logic card</i> , page 12 - 19
2:004	Backup copies do not match. Backup copy in secondary unit (T1) does not match data in primary unit (A5) Not influenced	- Hardware (T1 or A5) has been replaced. - Software has been updated to an older version - Corrupt memory T1 - Corrupt memory A5	1) Copy the truck configuration using TruckCom See the separate manual for TruckCom. 2) Copy the truck configuration using TruckCom See the separate manual for TruckCom. 3) Replace T1 See section "14.7.2 Replacing the motor control". 4) Replace A5 See section "13.1.6 Replacing the logic cards".
2:005	Internal program error Internal program error Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Internal program error A5	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
2:006	Saving to memory failed. Saving to memory failed because low battery voltage. Battery voltage below 15 V at the time of saving Truck behaviour Not influenced	- Battery connector disconnected from truck that is started - Voltage loss on supply to A5 - The battery is defective - Battery parameter #107 has an erroneous value	1) Connect the battery connector. 2) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50 3) Inspect the battery.

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2:006	Saving to memory failed. Saving to memory failed because low battery voltage. Battery voltage below 15 V at the time of saving Truck behaviour Not influenced		4) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
2:007	Internal program error A5 Internal program error A5 Truck behaviour Not influenced	Internal program error	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
2:008	Internal program error A5 Internal program error A5 Truck behaviour Not influenced	Internal program error	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
2:009	Internal program error A5 Internal program error A5 Truck behaviour Not influenced	Internal program error	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
2:010	CAN communication warning A5 has received too many erroneous signals Truck behaviour Not influenced	- Error on CAN bus wiring or connections - Loose connections - CAN node defective - Software failure	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) Check for loose connections Switch on the truck, and pull carefully in the branches of the wiring harness to find any loose connections, and note when/if the error code appears. 3) Check the CAN node Deactivate and disconnect the option unit connected to the CAN wiring

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2:010	CAN communication warning A5 has received too many erroneous signals Truck behaviour Not influenced		(e.g. K110, B90, T14 and A36/A37). Activate and then connect them one by one to identify which unit produces the communication error. 4) Update the truck software Update the truck software package to the latest available version.
2:011	New software. New software installed. In order to look in the error code log to see when new software was installed. Truck behaviour Not influenced	New software installed	1) Logged in the error log to see when the software was updated
2:080	Internal program error Internal program error Truck behaviour Creep speed, blocked lifting/lowering movement	- Internal program error - Manufacturing defect in A5	1) Create a truck report. Send a report to the manufacturer. Update the truck software to the latest version. 2) Replace A5 See section <i>Removing the logic card</i> , page 12 - 19
2:102	Configuration error. Truck configuration does not match parameter 1102 Truck behaviour Creep speed, blocked lifting/lowering movement	Incorrect value for parameter 1102	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
2:180	Internal program error Internal program error Truck behaviour Creep speed, blocked lifting/lowering movement	Internal program error	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
2:181	Fault in height measuring Truck behaviour Creep speed	Logic error A5	1) Let the forks pass 1.8 m one time
2:182	Missing signal from the 1.8 m sensor Truck behaviour Creep speed, blocked lifting/lowering movement	Defective component B129	1) Check the height sensor B129 Check that B129 is securely fastened. Check that the magnet is in place. If not, install a new one. If the magnet is in place:

No.	Description	Felorsak(er)	Action
2:182	Missing signal from the 1.8 m sensor Truck behaviour Creep speed, blocked lifting/ lowering movement		Check that the lateral distance and the distance between the sensor and the magnet is OK. Perform the test of B129 TruckCom or Service Information mode 9. Low signal = actuated sensor; High signal = unactuated sensor
2:183	Missing signal B129 Expected signal from B129 not received during lifting or lowering movement Truck behaviour Creep speed, blocked lifting/ lowering movement	Defective component B129	1) Check the height sensor B129 Check that B129 is securely fastened. Check that the magnet is in place. If not, install a new one. If the magnet is in place: Check that the lateral distance and the distance between the sensor and the magnet is OK. Perform the test of B129 TruckCom or Service Information mode 9. Low signal = actuated sensor; High signal = unactuated sensor
2:184	Signal from B129 Signal from B129 despite open main contactor No test signal from B129 to T1 when it is energised. Truck behaviour Creep speed, blocked lifting/ lowering movement	- Defective component B129 - Broken wiring	1) Check the height sensor B129 Check that B129 is securely fastened. Check that the magnet is in place. If not, install a new one. If the magnet is in place: Check that the lateral distance and the distance between the sensor and the magnet is OK. Perform the test of B129 TruckCom or Service Information mode 9. Low signal = actuated sensor; High signal = unactuated sensor 2) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage with the functions activated

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No.	Description	Felorsak(er)	Action
2:281	Erroneous combination of signals Incorrect signal combination from platform position sensor. Truck behaviour Creep speed	- High signal from both B119 and B120 - Constantly high signal at inputs (T1:101) and (T1:97)	1) Check the sensor signal Check that the distance between the sensor and the opposite part is correct. Check that there is no dirt or foreign objects on the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct 2) Check the signal to the unit Compare the signals from the sensor with the input signal (use a volt meter and verify using TruckCom or the built-in test that the signals are the same) Check the pins on the unit and in the connector Replace the unit if you cannot get it to work properly
2:282	Platform sensor activated The platform sensor has been activated for more than 5 minutes without any truck activity. Truck behaviour Creep speed	- Platform loaded - Constantly high signal from B119 - Constantly high signal at inputs T1:101	1) Handling error - no action. 2) Check the sensor signal Check that the distance between the sensor and the opposite part is correct. Check that there is no dirt or foreign objects on the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct 3) Check the signal to the unit Compare the signals from the sensor with the input signal (use a volt meter and verify using TruckCom or the built-in test that the signals are the same) Check the pins on the unit and in the connector

No.	Description	Felorsak(er)	Action
2:282	Platform sensor activated The platform sensor has been activated for more than 5 minutes without any truck activity. Truck behaviour Creep speed		Replace the unit if you cannot get it to work properly
2:284	Erroneous combination of signals Incorrect signal combination from the gate position sensor Truck behaviour Creep speed	- High signal from both B121 and B122 - Constantly high signal at inputs (T1:88) and (T1:96)	1) Check the sensor signal Check that the distance between the sensor and the opposite part is correct. Check that there is no dirt or foreign objects on the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct 2) Check the signal to the unit Compare the signals from the sensor with the input signal (use a volt meter and verify using TruckCom or the built-in test that the signals are the same) Check the pins on the unit and in the connector Replace the unit if you cannot get it to work properly 3)
2:285	Incorrect signal combination from the support arm position sensor Truck behaviour Creep speed, blocked lifting/lowering movement	- High signal from both B62 and B65 - Constantly high signal at inputs (T1:95) and (T1:111)	1) Check the sensor signal Check that the distance between the sensor and the opposite part is correct. Check that there is no dirt or foreign objects on the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct 2) Check the signal to the unit Compare the signals from the sensor with the

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No.	Description	Felorsak(er)	Action
2:285	Incorrect signal combination from the support arm position sensor Truck behaviour Creep speed, blocked lifting/ lowering movement		input signal (use a volt meter and verify using TruckCom or the built-in test that the signals are the same) Check the pins on the unit and in the connector Replace the unit if you cannot get it to work properly
2:501	Low battery voltage Low battery voltage below 16.8 Volts for more than 2 seconds Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- The battery is discharged - Voltage loss on supply to A5 - The battery is defective - Battery parameter #107 has an erroneous value	1) Charge the battery. 2) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50 3) Inspect the battery. 4) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
2:502	Truck type not set Parameter 1001 is incorrectly set Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- A5 is replaced and/or erroneous value for factory parameters - Incorrect value for parameter 1001	1) Copy the truck configuration using TruckCom See the separate manual for TruckCom. 2) Check parameters. Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
2:503	Wrong truck model set. Parameter 1108 is wrongly set in combination with 1001 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- A5 is replaced and/or erroneous value for factory parameters - Wrong value for parameter 1108 and/or 1001	1) Copy the truck configuration using TruckCom See the separate manual for TruckCom. 2) Check parameters. Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.

No.	Description	Felorsak(er)	Action
2:504	Low battery voltage The battery voltage to the capacitors in T1 is below 16.0 Volt when the main contactor must be closed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- The battery is discharged - Voltage loss on pre-supply to T1 - The battery is defective - Battery parameter #107 has an erroneous value	1) Charge the battery 2) Check the voltages in T1 Check fuses F50 and F1 Check the voltage using a volt meter (Measuring point F50 - B-) and compare with the TruckCom value "Battery Voltage" (Node:ACT, Analogue I/O) (Limit values?) (TBD) Check the voltage using a volt meter (Measuring point F1 - B-) and compare with the TruckCom value "Battery Voltage" (Node:ACT, Analogue I/O) (Limit values?) If the contactor is open, there is several volts of difference -- normal. If the contactor is closed, voltage should match the point across $\pm XX$ Check the connectors, joints and splices Check that any extra equipment is connected in accordance with our recommendation. 3) Inspect the battery. 4) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
2:511	A5 has received too many erroneous signals A5 has received too many erroneous signals Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Error on CAN bus wiring or connection - Loose connections - CAN node defective - Software failure	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm

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No.	Description	Felorsak(er)	Action
2:511	A5 has received too many erroneous signals A5 has received too many erroneous signals Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		2) Check for loose connections Switch on the truck, and pull carefully in the branches of the wiring harness to find any loose connections, and note when/if the error code appears. 3) Check the CAN node Deactivate and disconnect the option unit connected to the CAN wiring (e.g. K110, B90, T14 and A36/A37). Activate and then connect them one by one to identify which unit produces the communication error. 4) Update the truck software Update the truck software to the latest version.
2:515	No response is received from (DHU/TWIS) No return signal from K110 (DHU/TWIS) within 0.6 seconds after logging on Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Error on CAN bus wiring or connection • Wrong setting for parameter 102 • No connection with K110 • Loose connections • Software failure	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) Check parameters. Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 3) Check K110 (DHU/TWIS) Check that the DHU's green LED is lit. If the LED is not lit: Check the DHU's feed voltage = battery voltage Check that it is possible to connect TruckCom to the DHU. If it is possible

No.	Description	Felorsak(er)	Action
2:515	No response is received from (DHU/TWIS) No return signal from K110 (DHU/TWIS) within 0.6 seconds after logging on Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		to connect them, check the DHU's configuration. If there is a power supply but no communication, check the CAN bus. If it still not possible to communicate with the DHU, replace the unit. 4) Check for loose connections Switch on the truck, and pull carefully in the branches of the wiring harness to find any loose connections, and note when/if the error code appears. 5) Update the truck software Update the truck software to the latest version.
2:516	A5 has an incorrect box address. There is a jumper/short circuit in contact X203 between pin 1 and 0 V internally on the A5 card. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Box address error on A5	1) Check whether contact X203 is short circuited. If a jumper is fitted, remove it. 2) Replace A5 See section "13.1.6 Replacing the logic cards".
2:520	Parameter values do not match The checksum of the parameter values does not match the parameter values. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- New software has been loaded in the truck, adding a new parameter, or it has changed the parameter limit values - Corrupt memory circuit - Checksum error	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 2) Replace A5 See section "13.1.6 Replacing the logic cards". 3) Set new checksum If the problem with the check sum occurs directly in connection with the software update, proceed as follows: Change the value of one parameter in each category, operator parameters, service parameters and calibration parameters. Save the changes and restart the truck. Reset the values you

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2:520	Parameter values do not match The checksum of the parameter values does not match the parameter values. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		changed in the previous step back to their original values. Restart the truck. The program will have performed a new checksum calculation.
2:522	Incorrect power supply to the Hall sensors. High (or low) signal from all Hall sensors A5:S300-S318 Truck behaviour Plug braking to stop, then the parking brake is activated and the main contactor opens	Incorrect power supply to the Hall sensors.	1) Check the Hall sensors Check with TruckCom or the built-in test that the Hall sensors work the way they should. A Hall sensor that is being influenced should give a low signal. Replace A5 if the Hall sensors have an error function. See section "13.1.6 Replacing the logic cards".
2:523	Internal memory (FRAM or RAM) is faulty Reading from or writing to memory failed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Corrupt memory circuit	1) Replace A5 See section "13.1.6 Replacing the logic cards".
2:525	Truck software A5 does not have any truck software Truck behaviour The truck cannot be started	Software download interrupted	1) Update the truck software. Update the truck entire software package to the latest available version.
2:540	Contactor points in the main contactor fused High voltage over the capacitor bank internally in the motor control unit on the secondary side of the connector when it is open Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Contactor points in the main contactor are fused. Q10:3, Q10:4	1) Check contactor Check the function of the main contactor and that there is no connection between the points. Check that there is no short-circuit that can cause improper power supply. If this voltage does not increase when the main contactor pulls and the voltage is close to battery voltage (measured via precharge in), it is a sign that the contactor points in the main contactor are fused. Replace the contactor.

No.	Description	Felorsak(er)	Action
2:541	Contactor points fused Contactor points for steering servo and pump etc. in the main contactor are fused. Voltage on this side is measured through the steering servo. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Contactor points in the main contactor are fused. Q10:1, Q10:2	1) Check contactor Check the function of the main contactor and that there is no connection between the points. Check that there is no short-circuit that can cause improper power supply. On this side of the main contactor, there is no PTC resistor that can precharge any capacitor bank. So, this voltage drops towards zero when the main contactor is open. Check that the voltage drops at least 1 V over 5 seconds before trying to pull the main contactor again. Replace the contactor.
2:560	Main contactor ordered to close but stays open. Low voltage on the secondary side of the contactor when it is closed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Fuse F1 defective - Poor or no connection between the points. - Broken wiring	1) Check fuses Check if there is any contact on the fuse. Replace the fuse if it is defective. 2) Check contactor Check the contactor's function and that there is a connection between the points [Q10:3] and [Q10:4]. Check that there is no short-circuit that can cause improper power supply. Replace the contactor. 3) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage for the activated functions
2:561	Main contactor ordered to close but stays open. Low voltage on the secondary side of the contactor when it is closed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Fuse F53 defective - Poor or no connection between the points. - Broken wiring	1) Check fuses Check if there is any contact on the fuse. Replace the fuse if it is defective. 2) Check contactor Check the contactor's function and that there is a connection between the points [Q10:1] and

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2:561	Main contactor ordered to close but stays open. Low voltage on the secondary side of the contactor when it is closed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		[Q10:2]. Check that there is no short-circuit that can cause improper power supply. Replace the contactor. 3) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage with the functions activated
2:580	Signal from Hall element The Hall element for the speed control gives a signal (active low) when it should not Truck behaviour Plug braking to stop, then the main contactor opens and the parking brake is activated	A5 defective	1) Check the Hall sensors Check with TruckCom or the built-in test for the Hall sensors that they work the way they should. A Hall sensor that is being influenced should give a low signal. Replace A5 if the Hall sensors have an error function. See section "13.1.6 Replacing the logic cards".
3:001	The speed control was not in the neutral position at start-up One or more Hall elements for the speed control give an active signal at start Truck behaviour The truck cannot be driven, other functions OK	- Control manipulated. - The speed control does not spring back to neutral position. - A5 defective	1) Let go of the control, and wait for four seconds 2) Check the speed control Check the spring function Check that the speed control is not seized Check that the plastic cover where the spring is attached is intact. 3) Check the Hall sensors Check with TruckCom or the built-in test for the Hall sensors that they work the way they should. A Hall sensor that is being influenced should give a low signal. Replace A5 if the Hall sensors have an error function. See section "13.1.6 Replacing the logic cards".

No.	Description	Felorsak(er)	Action
3:080	Temperature sensor out of range The temperature sensor in the motor control power stage is outside the limit values. Above 130°C Truck behaviour The motor control is limiting the voltage.	Cooling is reduced or the temperature sensor is defective.	1) Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1,T13 or replace B1 with a repair kit.
3:081	The motor temperature sensor is outside of limit values. Above 210°C. Truck behaviour Creep speed and motor control limits the voltage.	Cooling is reduced or the temperature sensor is defective.	1) Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature sig-

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No.	Description	Felorsak(er)	Action
3:081	The motor temperature sensor is outside of limit values. Above 210°C. Truck behaviour Creep speed and motor control limits the voltage.		nals are plausible relative to the real temperatures of the components. If not, replace T1,T13 or replace B1 with a repair kit.
3:100	Motor control warning. Unknown warning Truck behaviour Creep speed, blocked lifting/ lowering movement	- Internal program error - Software failure	1) Create a truck report. Send a report to the manufacturer. 2) Update the truck software to the latest version.
3:120	Incorrect checksum Incorrect checksum for the motor control parameters Truck behaviour Creep speed, blocked lifting/ lowering movement	Motor control corrupt	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:127	Output current The strength of the current through the horn is more than 200mA even though the output is deactivated Truck behaviour Not influenced	Defective transistor on the output	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:128	Output current The strength of the current through the output to the fans is more than 200mA even though the output is deactivated Truck behaviour Not influenced	Defective transistor on the output	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:140	Short circuit in horn output Hardware protection in T1 detects a short circuit in the output for the horn. The current through the horn is above the valid limit value. Maximum value = 500mA, continuous value = 300mA Truck behaviour Not influenced	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component

No.	Description	Felorsak(er)	Action
3:141	Short circuit in fan output. Hardware protection in T1 detects a short circuit in the output for the fans. The current through the fan is above the valid limit value. Maximum value = 2.5A, continuous value = 1.5A Truck behaviour Not influenced	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
3:225	High motor rpm T1 detects to high speed for the motor, more than 5000 rpm Truck behaviour Creep speed	Defective component	1) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted.
3:303	Motor control start-up failed Truck behaviour The truck cannot be started	- Error on CAN bus wiring or connection - Incorrect value for parameter 1001 - No power supply - Wrong kind of motor control	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) Check parameters. Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 3) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in

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3:303	Motor control start-up failed Truck behaviour The truck cannot be started		any wire. Measure and check the voltage for the activated functions 4) Replace T1 See section "14.7.2 Replacing the motor control".
3:500	Internal error in T1 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:502	Error in T1 Error in T1 when saving parameters Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:505	Unknown error in T1 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• No software compatibility	1) Update the truck software Update the truck entire software package to the latest available version.
3:506	Low battery voltage Motor control detects excessively low battery voltage at start-up. Battery voltage below 15.3V after 10 seconds from power-on Truck behaviour The truck cannot be started	• Battery discharged • Voltage loss on supply to T1	1) Charge the battery 2) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50
3:507	Internal error in T1 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".

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3:511	CAN communication problem CAN communication problem between T1 and A5, A5 gets no signals from T1 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Error on CAN bus wiring or connection - CAN module in A5 defective	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) Replace the A5, see section "13.1.6 Replacing the logic cards".
3:512	CAN communication problem CAN communication problem between A5 and T1, T1 gets no signals from A5 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Error on CAN bus wiring or connection	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
3:514	CAN communication problem CAN communication problem, A5 cannot send to T1 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Error on CAN bus wiring or connection - CAN module in A5 defective	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) Replace A5 See section "13.1.6 Replacing the logic cards".

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No.	Description	Felorsak(er)	Action
3:520	Signal earthing in T1 Signal earthing in T1 out of valid range (TBD) Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	The ground for speed sensor and temperature sensor is connected to chassis, battery- or battery+	
3:523	Output current The amperage through the output to the brake (Q1) is more than 200mA even though the output is deactivated Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:524	Output current The amperage through the output to the main contactor (Q10) is more than 200mA even though the output is deactivated Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:525	High motor rpm T1 detects to high speed for the motor, more than 6000 rpm Truck behaviour The drive is stopped immediately with the parking brake	Defective component	1) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted.
3:540	Short circuit in power stage Hardware in T1 detects a short circuit in the power stage and a phase-to-phase short circuit, phase B- and phase B+ Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Short circuit in motor - Short circuit in motor wiring harness - Short circuit in power stage	1) Check the motor Disconnect motor wiring including sensor wires. Insulation test. Check that the coils are not electrically connected to the chassis. Replace M1 if the motor is not isolated 2) Check the motor wiring harness Visual check of the motor wiring harness. Check that there is no contact between the pole connections on the motor/motor control. Search for contact and damaged insulation on the wiring

No.	Description	Felorsak(er)	Action
3:540	Short circuit in power stage Hardware in T1 detects a short circuit in the power stage and a phase-to-phase short circuit, phase B- and phase B+ Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		harness. Disconnect the motor wiring harness at the motor control and the motor and measure the ohms of the wiring harness between the phases and verify that there is no contact. 3) Replace T1 See section "14.7.2 Replacing the motor control".
3:542	High current in speed sensor Too many amperes to the speed sensor, limit value 68 mA Truck behaviour The drive is stopped immediately with the parking brake	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
3:545	Output short circuit Hardware protection in T1 detects a short circuit in the output to the brake (Q1) The current through the brake exceeds the limit values. Maximum value = 5.95A, continuous value = 3A Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component

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No.	Description	Felorsak(er)	Action
3:546	Short circuit or current in main contactor Hardware protection in T1 detects a short circuit or current through the main contactor (Q10) exceeds valid limit values. Maximum value = 2.5A, continuous value = 1.5A Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
3:547	Output short circuit Hardware protection in T1 detects a short circuit in the output "High Side driver" Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Components are using too much power or wiring is short circuited.	
3:560	Low current The amperage to speed sensor (B11) is below the 6mA limit value Truck behaviour The drive is stopped immediately with the parking brake	- Broken wiring - Defective component	1) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage for the activated functions 2) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted.
3:563	Low current The amperage is below the limit value 100mA on the output to brake (Q1) Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component

No.	Description	Felorsak(er)	Action
3:563	Low current The amperage is below the limit value 100mA on the output to brake (Q1) Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		ent, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
3:564	Low current The amperage is below the limit value 100mA on the output to brake (Q10) Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
3:584	Power supply outside of valid range Power supply to B11 out of valid range. External 12V outside of valid range: 10V - 14V Truck behaviour The drive is stopped immediately with the parking brake	- High power consumption or short circuit. - Defective motor control.	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component 2) Replace T1 See section "14.7.2 Replacing the motor control".

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No.	Description	Felorsak(er)	Action
3:586	High current Software in T1 detects too high a current in the power stage and also high current phase-phase, phase B- and phase B+. The current is 150% of maximum current. Truck behaviour The drive is stopped immediately with the parking brake	• Short circuit in motor • Short circuit in motor wiring harness • Short circuit in power stage	1) Check the motor Disconnect motor wiring including sensor wires. Insulation test. Check that the coils are not electrically connected to the chassis. Replace M1 if the motor is not isolated 2) Check the motor wiring harness Visual check of the motor wiring harness. Check that there is no contact between the pole connections on the motor/motor control. Search for contact and damaged insulation on the wiring harness. Disconnect the motor wiring harness at the motor control and the motor and measure the ohms of the wiring harness between the phases and verify that there is no contact. 3) Replace T1 See section "14.7.2 Replacing the motor control".
3:587	Battery voltage exceeds 33 V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Voltage loss on supply to T1 • Battery defective	1) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50 2) Inspect the battery.
3:589	Battery voltage is below 12 V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• The battery is discharged • Voltage loss on power supply to T1 • The battery is defective • Battery parameter # 107 has an erroneous value	1) Charge the battery. 2) Check the voltages in T1 Check fuses F50 and F1 Check the voltage using a volt meter (Measuring point F50 - B-) and compare with the TruckCom value "Battery Voltage" (Node:ACT, Analogue I/O) (Limit values?) (TBD) Check the voltage using a volt meter (Measuring

No.	Description	Felorsak(er)	Action
3:589	Battery voltage is below 12 V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		point F1 - B-) and compare with the TruckCom value "Battery Voltage" (Node:ACT, Analogue I/O) (Limit values?) If the contactor is open, there is several volts of difference -- normal. If the contactor is closed, voltage should match the point across $\pm XX$ Check the connectors, joints and splices Check that any extra equipment is connected in accordance with our recommendation. 3) Inspect the battery. 4) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
3:590	Change in rpms The change exceeds 1000 rpm in 20 ms Truck behaviour The drive is stopped immediately with the parking brake	• Broken wiring • Defective component • Change in surface friction combined with slippage	1) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage with the functions activated 2) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted. 3) Drive carefully if there is a risk of slipping
3:591	Improbable motor rpm Rpms over 15.000 rpm Truck behaviour The drive is stopped immediately with the parking brake	• Defective component	1) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed

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No.	Description	Felorsak(er)	Action
3:591	Improbable motor rpm Rpm's over 15.000 rpm Truck behaviour The drive is stopped immediately with the parking brake		wheel on the motor axle is clean, intact and properly mounted.
3:592	Internal electrical fault. Sensor offset is too high Truck behaviour The drive is stopped immediately with the parking brake	Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:593	Signal error B11. Loss of channel A in the rotational speed sensor detected Truck behaviour The drive is stopped immediately with the parking brake	- Broken wiring - Defective component	1) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage with the functions activated 2) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted.
3:594	Signal error B11. Loss of channel B in the rotational speed sensor detected Truck behaviour The drive is stopped immediately with the parking brake	- Broken wiring - Defective component	1) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage with the functions activated 2) Check the rotational speed sensor Disconnect the B11. Check that there are no metal fragments on B11 sensor points. Check that the toothed wheel on the motor axle is clean, intact and properly mounted.
3:596	High temperature The motor control unit temperature exceeds +110°C. Truck behaviour The drive is stopped immediately with the parking brake	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor

No.	Description	Felorsak(er)	Action
3:596	High temperature The motor control unit temperature exceeds +110°C. Truck behaviour The drive is stopped immediately with the parking brake		regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
3:597	High temperature The motor temperature exceeds +180°C. Truck behaviour The drive is stopped immediately with the parking brake	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature sig-

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No.	Description	Felorsak(er)	Action
3:597	High temperature The motor temperature exceeds +180°C. Truck behaviour The drive is stopped immediately with the parking brake		nals are plausible relative to the real temperatures of the components. If not, replace T1,T13 or replace B1 with a repair kit.
3:598	Internal voltage is outside of permitted range Internal 5V outside of valid range: 4.12V - 5.88V or internal 15V voltage outside valid range: 10.56V - 19V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
3:599	Voltage regulator for analogue sensors Outside of valid range: 4V - 6V Truck behaviour Creep speed, blocked lifting/lowering movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component.
4:001	Lifting/lowering control is not in neutral position at start-up. Lifting is prevented until the lifting/lowering controller is back in the neutral position. Truck behaviour Blocked lifting/lowering movement	Control manipulated	1) Let go of the control, and wait for four seconds
4:003	Armature current on pump motor too low. Check F2 Truck behaviour Blocked lifting movement	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and re-

No.	Description	Felorsak(er)	Action
4:003	Armature current on pump motor too low. Check F2 Truck behaviour Blocked lifting movement		place the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:101	Signal error The signal from the sensilift is out of valid range: 0.23V - 4.71V Truck behaviour Blocked lifting/lowering movement	Defective component, short circuit or wire cut off	1) Check the sensilift Check that the sensor has a supply voltage of approx. 5V Compare the measured output from the sensilift with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensilift potentiometer or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in A5. - Replace A5. See section "13.1.3 Replacing the logic cards".
4:102	Signal error Signal from the sensilift potentiometer detected even though parameter 1102 does not have the value 4. Truck behaviour Blocked lifting/lowering movement	- Incorrect value for parameter 1102 - Error in hydraulic controls	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 2) Replacement of the hydraulic control. See section "13.1 Steering arm C4119".
4:103	Internal error in T1 Truck behaviour Blocked lifting movement	Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:105	Voltage from B127 is outside of permitted range: 0.3 V - 4.7 V Truck behaviour Creep speed, blocked lifting movement	- Defective component, short circuit or wire cut off - Tilt sensor in the wrong position - Tilt sensor incorrectly calibrated	1) Check the analogue sensors Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it in-

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
4:105	Voltage from B127 is outside of permitted range: 0.3 V - 4.7 V Truck behaviour Creep speed, blocked lifting movement		icates an error in the sensor or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace T1 See section "14.7.2 Replacing the motor control". 2) Check the tilt sensor Check that B127 has been fitted correctly and securely. Make sure the truck is parked on a level surface. Check the signal from B127 via Service Information mode 8 or Truck-Com. Type value 2.5 V = 0°. Adjust B127 mechanically if necessary. Calibrate B127 via calibration 4 3) See point 2
4:107	Valve does not close Expected position of valve Q71 does not match read position of B123. Decompression lock valve Q71 is stuck in the open position Truck behaviour Creep speed, blocked lifting movement	B123 incorrectly adjusted	1) Check decompression locking
4:108	B140 indicates a low oil level in spring decompression locking Truck behaviour Creep speed, blocked lifting movement		1) Check decompression locking
4:110	Template Expected position of valve Q71 does not match read position of B123. Decompression lock valve Q71 is stuck in the closed position Truck behaviour Creep speed, blocked lifting movement	B123 incorrectly adjusted	1) Check decompression locking

No.	Description	Felorsak(er)	Action
4:120	Sensor offset is too high Truck behaviour Blocked lifting movement	Defective motor control	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:121	High current Software in T1 detects too high a current in the pump power stage The current is 150% of maximum current. Truck behaviour Blocked lifting movement	Pump motor or its wiring harness defective	1) Check the pump motor Disconnect the motor cables. Insulation test. Check that the coils are not electrically connected to the chassis. If motor is not insulated, replace the motor (M3)
4:123	Field winding output overloaded Truck behaviour Blocked lifting movement	- Intensive use of the truck - Pump motor or its wiring harness defective	1) Intensive use of the truck. Let the truck cool down. 2) Check the pump motor Disconnect the motor cables Insulation test. Check that the coils are not electrically connected to the chassis. If motor is not insulated, replace the motor (M3)
4:180	Pressure sensor signal B4 is outside the valid range: 0.3 V - 4.86 V Truck behaviour Creep speed, blocked lifting movement	Defective component, short circuit or wire cut off	1) Check the analogue sensors Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensor or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace T1 see section "14.7.2 Replacing the motor control".
4:181	Signal outside of valid range The signal from the pressure sensor B5 is out of valid range: 0.3 V - 4.86 V Truck behaviour Creep speed, blocked lifting movement	Defective component, short circuit or wire cut off	1) Check the analogue sensors Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding

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No.	Description	Felorsak(er)	Action
4:181	Signal outside of valid range The signal from the pressure sensor B5 is out of valid range: 0.3 V - 4.86 V Truck behaviour Creep speed, blocked lifting movement		value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensor or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace T1 See section "14.7.2 Replacing the motor control".
4:200	Low pressure in Powertrak circuit during transport (lifting height < 1.8 m). Signal from B5 too low in relation to B4. Min. value 300 mV Truck behaviour Creep speed, blocked lifting movement	- Defective component, short circuit or wire cut off - Powertrak	1) Check the analogue sensors Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensor or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace T1 See section "14.7.2 Replacing the motor control". 2) Check PowerTrak
4:201	High pressure in Powertrak circuit in stacking mode (lifting height > 1.8 m). Signal from B5 too high. Max. value 1500 mV Truck behaviour Creep speed, blocked lifting movement	- Defective component, short circuit or wire cut off - Powertrak	1) Check the analogue sensors Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensor or the wiring harness. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace T1

No.	Description	Felorsak(er)	Action
4:201	High pressure in Powertrak circuit in stacking mode (lifting height > 1.8 m). Signal from B5 too high. Max. value 1500 mV Truck behaviour Creep speed, blocked lifting movement		See section "14.7.2 Replacing the motor control". 2) Check PowerTrak
4:231	Current through output to Powertrak valve (Q58) is more than 200 mA even though the output is not active. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:232	Output current Exceeds 200mA to the pump contactor (Q25) when it is not active. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:233	Current through output to Q70 Exceeds 200mA even though the output is not active Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:234	Current through output to Q4/Q40 Current through the Q4/Q40 output is more than 200 mA even though the output is inactive. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:235	Current through output to Q41/Q23 Exceeds 200 mA even though the output is not active Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".

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No.	Description	Felorsak(er)	Action
4:236	Too high current in output to Q59 Current more than 200mA when not activated Truck behaviour Plug braking to stop, then the parking brake is activated and the main contactor opens	Defective transistor on the output.	1) Replace T1 See section "14.7.2 Replacing the motor control".
4:238	Template Truck behaviour Plug braking to stop, then the parking brake is activated and the main contactor opens		1) 2) 3)
4:242	Pump motor (M3) output is short circuited or too high a current through the output Truck behaviour Plug braking to stop, then the parking brake is activated and the main contactor opens	Pump motor or its wiring harness defective	1) Check the pump motor Disconnect the motor cables Insulation test. Check that the coils are not electrically connected to the chassis. If motor is not insulated, replace the motor (M3)
4:243	Lowering valve (Q4/Q40) Output is short circuited or current too high through the output to lowering valve. Valid limit values - maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Blocked lifting/lowering movement	High power consumption or short circuit.	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
4:244	Solenoid (Q41/Q23) Output is short circuited or current too high through the output to lowering valve. Valid limit values - maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Blocked lifting/lowering movement	High power consumption or short circuit.	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is

No.	Description	Felorsak(er)	Action
4:244	Solenoid (Q41/Q23) Output is short circuited or current too high through the output to lowering valve. Valid limit values - maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Blocked lifting/lowering movement		jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
4:245	Drain valve Powertrak (Q58) Output is short circuited or current too high through the output to the drain valve. Valid limit values - maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Creep speed, blocked lifting movement	High power consumption or short circuit.	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
4:246	Hardware protection in T1 Detected a short circuit in output Q25 (pump contactor). The current through the contactor (Q25) is above the valid limit value. Maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Blocked lifting/lowering movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component

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No.	Description	Felorsak(er)	Action
4:247	Hardware protection in T1 Detected a short circuit in output Q70 (decompression locking valve) or current through the valve output exceeds the limit values. Maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Creep speed, blocked lifting movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
4:248	Filling valve (Powertrak Q59) Detected a short circuit in output Q59 or current through the valve output exceeds the limit values. Maximum value = 2.5 A, continuous value = 1.5 A Truck behaviour Creep speed, blocked lifting movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
4:250	Short circuit Short circuit detected Truck behaviour Creep speed, blocked lifting movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component

No.	Description	Felorsak(er)	Action
4:280	High temperature The temperature sensor for the pump power stage in T1 is outside of the limit values. The temperature shown exceeds 130°C Truck behaviour Blocked lifting movement	Cooling is reduced or the temperature sensor is defective.	1) Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
4:281	High temperature The temperature in T1's pump power stage exceeds 85°C. Truck behaviour Blocked lifting movement	Cooling is reduced or the temperature sensor is defective.	1) Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature sig-

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No.	Description	Felorsak(er)	Action
4:281	High temperature The temperature in T1's pump power stage exceeds 85°C. Truck behaviour Blocked lifting movement		nals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
4:561	Current on Powertrak valve (Q58) output is too low Current below limit of 100mA Truck behaviour Creep speed, blocked lifting movement	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:562	Pump contactor output current too low (Q22, Q25, Q52) Limit value 100mA Truck behaviour Blocked lifting/lowering movement	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:563	Current on decompression locking valve (Q70) output is too low Limit value 100mA Truck behaviour Creep speed, blocked lifting movement	Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are

No.	Description	Felorsak(er)	Action
4:563	Current on decompression locking valve (Q70) output is too low Limit value 100mA Truck behaviour Creep speed, blocked lifting movement		outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:564	Low current Too low a current on lowering valve output (Q4 - Q40) - limit value 100 mA Truck behaviour Blocked lifting/lowering movement	• Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:565	Too low a current on valve output (Q23 - Q41) Limit value 100mA Truck behaviour Blocked lifting/lowering movement	• Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:566	Current on filler valve (Powertrak) output is too low Truck behaviour Creep speed, blocked lifting movement	• Low power consumption or break	1) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
4:566	Current on filler valve (Powertrak) output is too low Truck behaviour Creep speed, blocked lifting movement		outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions.
4:568	Template Truck behaviour Creep speed, blocked lifting movement	High power consumption or short circuit	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component
5:035	High temperature The temperature in the steering motor power stage exceeds 68°C. Truck behaviour Not influenced	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
5:035	High temperature The temperature in the steering motor power stage exceeds 68°C. Truck behaviour Not influenced		If not, replace T1,T13 or replace B1 with a repair kit.
5:080	Reference sensor error Current below limit of 500µA Truck behaviour Not influenced	- Broken wiring - Defective component	1) Check wiring after the break. Disconnect the battery Using instruments, check that there is no break in any wire. Measure and check the voltage for the activated functions 2) Check the reference sensor Steer slightly to the right so that the gear ring is correctly positioned in relation to the reference sensor. Cut off power to the truck. Check the clearance between the sensor and the gear ring elevation using a feeler gauge. Nominal measurement: 1.5 mm Note: Risk of personal injury if power is restored to the truck while work is performed on the gear ring. Restore power to the truck. Check if there is any power supply to B17, 9.8 - 13.5V between wires 95 and 97. If there is a power supply, replace B17. Check if there is a short circuit between wires 95 and 97, but also between wire 95 and the rest of the electrical system. If there is no power supply or short circuit, replace the T13
5:081	Power supply to reference sensor outside of valid range. It should be 9.8 - 13.5V between the wires 95-97. Truck behaviour Not influenced	Defective component	1) Check the reference sensor Steer slightly to the right so that the gear ring is correctly positioned in relation to the reference sensor.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
5:081	Power supply to reference sensor outside of valid range. It should be 9.8 - 13.5V between the wires 95-97. Truck behaviour Not influenced		Cut off power to the truck. Check the clearance between the sensor and the gear ring elevation using a feeler gauge. Nominal measurement: 1.5 mm Note: Risk of personal injury if power is restored to the truck while work is performed on the gear ring. Restore power to the truck. Check if there is any power supply to B17, 9.8 - 13.5V between wires 95 and 97. If there is a power supply, replace B17. Check if there is a short circuit between wires 95 and 97, but also between wire 95 and the rest of the electrical system. If there is no power supply or short circuit, replace the T13
5:135	Steering motor overloaded Overloaded steering motor to 80% of maximum capacity. Truck behaviour Creep speed	- Intensive steering with stationary truck - Turning of drive wheel was prevented.	1) Avoid steering the truck when stationary 2) Checking the steerability of the drive unit. Check why the drive wheel cannot be turned with the help of the steering motor in the following places. Steering motor Gears of the steering motor Gears Steering bearings Crown wheel/ steering motor gear Drive wheel/surface
5:136	Error in power supply to brake coil in (A83). Current differs from requested current by 100 mA and current exceeds 2 A. Truck behaviour Creep speed	- High power consumption or short circuit - Low power consumption or break	1) Check for overcurrent or short circuit: Check the ohms and the amperes of the component, compare them to the setpoint values if they outside the accepted range. Check the connections and wiring harness for short-circuit. Check that the component does not seize or is

No.	Description	Felorsak(er)	Action
5:136	Error in power supply to brake coil in (A83). Current differs from requested current by 100 mA and current exceeds 2 A. Truck behaviour Creep speed		jammed. If points 1-3 are OK, it indicates that the output stage is short circuited - Replace the power-supplying component. 2) Check wiring and component after the break Disconnect the battery Visually inspect the wiring harness Using instruments, check that there is no break in any wire Check the Ohms and the Amperes of the component, compare them to the setpoint values, and replace the unit if they are outside the accepted range. Reconnect the battery. Measure and check the voltage for the activated functions. 3)
5:180	Steering angle signal is outside the valid range (applies only to mechanical steering, not power steering) Valid range: 0.18V - 4.86V Truck behaviour Creep speed	Defective component, short circuit or wire cut off	1) Check the analogue sensor Check that the sensor has a 5V or 24V power supply (depending on the component) Compare the measured output from the sensor with the corresponding value in TruckCom or built-in test. If the values are the same and are out of the valid range, it indicates an error in the sensor or the wiring. Repair or replace the component. If the values are not the same, it points to an error in T1. Replace the T1. See section "14.7.2 Replacing the motor control".
5:182	When B17 indicates 0 degrees, A83 must lie in the interval +/- 7 degrees Truck behaviour Creep speed	- The clearance between reference sensor and crown wheel is wrong - Defective component - Erroneous steering angle signal	1) Check the reference sensor Steer slightly to the right so that the gear ring is correctly positioned in relation to the reference sensor. Cut off power to the

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No.	Description	Felorsak(er)	Action
5:182	When B17 indicates 0 degrees, A83 must lie in the interval +/- 7 degrees Truck behaviour Creep speed		truck. Check the clearance between the sensor and the gear ring elevation using a feeler gauge. Nominal measurement: 1.5 mm Note: Risk of personal injury if power is restored to the truck while work is performed on the gear ring. Restore power to the truck. Check if there is any power supply to B17, 9.8 - 13.5V between wires 95 and 97. If there is a power supply, replace B17. Check if there is a short circuit between wires 95 and 97, but also between wire 95 and the rest of the electrical system. If there is no power supply or short circuit, replace the T13 2) See above 3) Check the steering angle sensor Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Digital PWM signal from A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration.

No.	Description	Felorsak(er)	Action
5:301	Steering motor control power stage defective Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Voltage loss on supply to T13 - No software compatibility - Defective electronic unit	1) Check the voltages in T13 Check that there is battery voltage between conductors 23/[A5:8] + [A5:16] and 40/[A5:7] + [A5:15]. Check fuse F53 2) Update the truck software Update the truck entire software package to the latest available version. 3) Replace the T13, see section "10.2.2 Replacing the steering motor".
5:303	Start of T13 failed Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Voltage loss on supply to T13 - Error on CAN bus wiring or connections - No software compatibility	1) Check the voltages in T13 Check the voltage using a volt meter (Measuring points wires 40 and 23 at T13), and compare with the value from Truck-Com "DC Bus Voltage" Compare with battery voltage. Check the connectors, joints and splices. Check fuse F53 2) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 3) Update the truck software Update the truck entire software package to the latest available version.
5:320	Voltage below valid level: 12V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Voltage loss on supply to T13 - The battery is defective - Battery parameter # 107 has an erroneous value	1) Check the voltages in T13 Check the voltage using a volt meter (Measuring points wires 40 and 23 at T13), and compare with

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
5:320	Voltage below valid level: 12V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		the value from Truck-Com "DC Bus Voltage" Compare with battery voltage. Check the connectors, joints and splices. Check fuse F53 2) Inspect the battery. 3) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
5:325	Overloaded steering motor Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Turning of drive wheel was prevented. - Intensive steering with stationary truck	1) Checking the steerability of the drive unit. Check why the drive wheel cannot be turned with the help of the steering motor in the following places. Steering motor Gears of the steering motor Gears Steering bearings Crown wheel/ steering motor gear Drive wheel/surface 2) Avoid steering the truck when stationary
5:326	High temperature The temperature in the steering motor power stage exceeds 85°C. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from

No.	Description	Felorsak(er)	Action
5:326	High temperature The temperature in the steering motor power stage exceeds 85°C. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
5:335	The internal parameters in the steering motor are erroneous Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- No software compatibility - Wrong electronic unit in the motor servo in connection with change - Defective electronic unit	1) Update the truck software Update the truck software to the latest version. 2) Check the P/N of the electronic unit. 3) Replace the T13, see section "10.2.2 Replacing the steering motor".
5:336	The internal parameters in the steering motor are erroneous Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- No software compatibility - Wrong electronic unit in the motor servo in connection with change - Defective electronic unit	1) Update the truck software Update the truck software to the latest version. 2) Check the P/N of the electronic unit. 3) Replace the T13, see section "10.2.2 Replacing the steering motor".
5:337	T13 reports an error during reference run Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Defective component - Turning of drive wheel was prevented.	1) Check the reference sensor Steer slightly to the right so that the gear ring is correctly positioned in relation to the reference sensor. Cut off power to the truck. Check the clearance between the sensor and the gear ring elevation using a feeler gauge. Nominal measurement: 1.5 mm Note: Risk of personal injury if power is restored to the truck while work is performed on the gear ring. Restore power to the truck. Check if there is any power supply to B17, 9.8 - 13.5V between wires 95 and 97.

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No.	Description	Felorsak(er)	Action
5:337	T13 reports an error during reference run Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		If there is a power supply, replace B17. Check if there is a short circuit between wires 95 and 97, but also between wire 95 and the rest of the electrical system. If there is no power supply or short circuit, replace the T13. See section "10.2.2 Replacing the steering motor". 2) Checking the steerability of the drive unit. Check why the drive wheel cannot be turned with the help of the steering motor in the following places. Steering motor Gears of the steering motor Gears Steering bearings Crown wheel/ steering motor gear Drive wheel/surface
5:380	High temperature Temperature in steering motor exceeds high temperature Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperature

No.	Description	Felorsak(er)	Action
5:380	High temperature Temperature in steering motor exceeds high temperature Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		ures of the components. If not, replace T1,T13 or replace B1 with a repair kit.
5:414	CAN communication problem Between A5 and T13. T13 does not get any commands from A5 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
5:415	CAN communication problem Between A5 and T13. A5 gets no responses from T13 or answers are corrupted Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
5:416	CAN communication problem Between A5 and T13. A5 gets no response from T13. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
5:417	CAN communication problem Between A5 and T13 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
5:421	Voltage above limit of 39V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Voltage loss on supply to T1 - Voltage measurement error in T13 - The battery is defective	1) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50 2) Check the voltages in T13 Check the voltage using a volt meter (Measuring points wires 40 and 23 at T13), and compare with the value from Truck-Com "DC Bus Voltage" Compare with battery voltage. Check the connectors, joints and splices. Check fuse F53 3) Inspect the battery.
5:422	Current exceeds limit value 95 A. Short circuit in steering motor power stage. Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Short circuit in steering motor - Short circuit in power stage	1) Replace M6 See RM for instructions (TBD) 2) Replace T13 See section "10.2.2 Replacing the steering motor".
5:435	Internal program error Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Software failure	1) Update the truck software to the latest version.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
5:436	Interruptions in one or more phases in T13/M6 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Interruption in servo motor • Interruption in power stage	1) Replace M6 See RM for instructions (TBD) 2) Replace T13 See RM for instructions (TBD)
5:437	T13 detects an erroneous signal from Hall sensor Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Defective components internally in M6 • Faulty input signal	1) Replace M6 See RM for instructions (TBD) 2) Replace T13 See RM for instructions (TBD)
5:438	T13 detects a problem with voltage measurement Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Internal error in electronics	1) Replace the T13 or the complete unit (TBD)
5:500	Unknown steering motor error Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• No software compatibility	1) Update the truck software Update the truck software to the latest version.
5:501	Internal program error Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Internal program error	1) Create a truck report. Send a report to the manufacturer.
5:502	A5 detects T13 Parameter 1107 is set to mechanical steering Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Incorrect value for parameter 1107	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
5:520	Power supply to A83 Outside of valid range: 4.5V-5.5V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	• Defective component, short circuit or internal error in T13	1) Check the power supply to the steering angle sensor Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Measure the voltage between wires 91 and 94. Set point: 4.5V - 5.5V

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No.	Description	Felorsak(er)	Action
5:520	Power supply to A83 Outside of valid range: 4.5V-5.5V Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		If the voltage is too high - replace T13 If the voltage is too low - open splice A83 and measure output 4 on T13. If the voltage is too low - replace T13, other- wise replace A83
5:540	T13 detects a short circuit between steering angle sensor signals 1 and 2 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective component, short circuit or internal error in T13	1) Check the steering angle sensor Check the wiring between T13 and A83 for short circuit or disrup- tion - repair or replace the wiring. Digital PWM signal from A83 cannot be meas- ured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibra- tion.
5:580	Reference run not terminated within 15 seconds Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Defective component - Turning of drive wheel was prevented	1) Check the reference sensor Steer slightly to the right so that the gear ring is correctly positioned in re- lation to the reference sensor. Cut off power to the truck. Check the clearance between the sensor and the gear ring elevation using a feeler gauge. Nominal measurement: 1.5 mm Note: Risk of personal injury if power is restored to the truck while work is performed on the gear ring. Restore power to the truck. Check if there is any power supply to B17, 9.8 - 13.5V between wires 95 and 97. If there is a power supply, replace B17. Check if there is a

No.	Description	Felorsak(er)	Action
5:580	Reference run not terminated within 15 seconds Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		short circuit between wires 95 and 97, but also between wire 95 and the rest of the electrical system. If there is no power supply or short circuit, replace the T13 2) Checking the steerability of the drive unit. Check why the drive wheel cannot be turned with the help of the steering motor in the following places. Steering motor Gears of the steering motor Gears Steering bearings Crown wheel/ steering motor gear Drive wheel/surface
5:582	Steering angle signal 1 and 2 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective component, short circuit or internal error in T13	1) Check the steering angle sensor Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Digital PWM signal from A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration.
5:583	Steering angle signal 1 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective component, short circuit or internal error in T13	1) Check the steering angle sensor Check steering angle signal 1 between pin 3 (contact A83) and pin 5 (contact T13) Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Digital PWM signal from

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No.	Description	Felorsak(er)	Action
5:583	Steering angle signal 1 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration.
5:584	Steering angle signal 2 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective component, short circuit or internal error in T13	1) Check the steering angle sensor Check steering angle signal 2 between pin 4 (contact A83) and pin 13 (contact T13) Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Digital PWM signal from A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration.
5:585	Incorrect configuration of steering motor Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	No software compatibility	1) Update the truck software Update the truck software to the latest version.
5:586	Sums of steering angle signals 1 and 2 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	Defective component, short circuit or internal error in T13	1) Check the steering angle sensor Check the wiring between T13 and A83 for short circuit or disruption - repair or replace the wiring. Digital PWM signal from

No.	Description	Felorsak(er)	Action
5:586	Sums of steering angle signals 1 and 2 out of valid range Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens		A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration.
6:300	Unknown auto error The autotruck system reports an error not explicitly mentioned in this list Truck behaviour The truck stops		1) Restart the truck
6:301	Auto mode timeout The autotruck system does not verify auto mode soon enough. Truck behaviour The truck stops	Software error in the autotruck system	1) Restart the truck 2) Make a truck report 3) Install the correct software in the auto system
6:302	Autotruck PDO error The Error active bit is set to 1 by the autotruck system 1= Is set as soon as there is an error code according to: (0x8250, 0x8102, 0x8103, 0x8104) and the truck is in auto mode. 0= Is set when the truck is used manually, or when the error code disappears. Truck behaviour The truck stops		Connect a computer to CVC600 Adapter wiring 1) Restart the truck 2) Make sure the CAN bus is intact all the way to CVC600.
6:310	Autotruck PDO error 0x8250 from the autotruck system CVC600 has no connection with node id 12. (Steering arm, Manual truck). Truck behaviour The truck stops		1) Restart the truck 2) Troubleshoot the CAN bus between CVC600 and the steering arm. 3) Also check if the steering arm has started correctly with the correct version.

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No.	Description	Felorsak(er)	Action
6:311	Autotruck SDIO error 0x8102 from the autotruck system CVC600 has no connection with node id 2. (Internal SDIO) Truck behaviour The truck stops		1) Restart the truck 2) Make sure the SDIO has the correct software version 3) Install the correct software
6:312	Lift encoder error 0x8103 from the autotruck system CVC600 has no connection with node id 5. (Lift encoder) Truck behaviour The truck stops		1) Restart the truck 2) Check the cable 3) Troubleshoot the CAN bus between CVC600 and the lift encoder. 4) Replace the height sensor: Link target missing for title: [ID: 2691305483]
6:313	Error in S100 scanner 0x8104 from the autotruck system CVC600 has no connection with node id 24. (S100 scanner) Truck behaviour The truck stops		1) Restart the truck 2) Troubleshoot the CAN bus between CVC600 and the S100 scanner. 3) Replace the S100 scanner: Link target missing for title: [ID: 2689806731]
6:314	Control panel OPT100 error 0x8105 from the autotruck system CVC600 has no connection with node id 3. (Opt100 Display) Truck behaviour The truck stops		1) Restart the truck 2) Troubleshoot the CAN bus between CVC600 and the S100 scanner. 3) Replace the control panel: Link target missing for title: [ID: 2700018827]
6:315	PDO transmission failed ICH failed to send CAN-PDO message to autotruck system. Truck behaviour The truck stops	- Large amount of high-priority messages on CAN bus, unit with faulty software connected to CAN. - CAN bus interference	1) Troubleshoot the CAN bus 2) Restart the truck 3) Make sure the correct software is installed in the autotruck system.
6:316	PDO reception failed No periodic CAN messages received by ICH from the autotruck system. Truck behaviour The truck stops	Interruption in the CAN bus, autotruck system error	1) Restart the truck 2) Troubleshoot the CAN bus 3) Make sure the correct software is installed in the autotruck system.
6:360	Parking brake connection error Too low voltage through the parking brake when it is activated Truck behaviour The truck stops	Brake circuit open	1) Troubleshoot connectors and wiring harness to the parking brake

No.	Description	Felorsak(er)	Action
6:360	Parking brake connection error Too low voltage through the parking brake when it is activated Truck behaviour The truck stops		2) Replace the drive encoder Link target missing for title: [ID: 2786376331]
6:382	Lift encoder error 0x7005 from the autotruck system The lifting monitoring has been triggered because the correct height was not reached within 3s. Truck behaviour The truck stops		1) Make sure nothing is blocking the forks. Make sure that the encoder is correctly adjusted. Link target missing for title: [ID: 2691314699]
6:383	Drive encoder error 0x7006 from the autotruck system The drive monitoring has been triggered because the correct speed was not reached within 1.5s. Truck behaviour The truck stops		1) Remove any rubbish around the drive wheel. 2) In some cases the support arm wheel may cause it to be difficult to change direction. It is then possible to add to the Integral part of the control to increase power to the motor.
6:384	Steering error 0x7007 from the autotruck system The steering monitoring has been triggered because the correct steering angle was not reached within 1s. Truck behaviour The truck stops		1) Restart the truck 2) If this happens in sharp corners in the layout, try to make these less abrupt. 3) Troubleshoot the wiring between T13 and A83 for a short circuit or disruption - repair or replace the wiring. Digital PWM signal from A83 cannot be measured, use an A83 that works as a reference. Compare the steering angle with the steering angle that is indicated by the system via Service Information Mode 13 or TruckCom - If it does not seem to be the same, check the mechanical fastening of the steering arm and make a new steering angle calibration. 4) Replace the steering angle sensor: <i>Replacing the steering angle sensor servo unit.</i> , page 12 - 71

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
6:385	Side guard error 0x7008 from the autotruck system The truck side sensors do not respond. Truck behaviour The truck stops		1) Check the side guard: Link target missing for title: [ID: 2691264395] 2) Calibrate the side sensor sensitivity: Link target missing for title: [ID: 2691261323] 3) Replace the side guard: Link target missing for title: [ID: 2691249035]
6:387	Safety relay error. The autotruck system reports error 0x7001 or 0x7003 Safety relay/Flexisoft is in the wrong position Truck behaviour The truck stops	- Flexisoft program error - Flexisoft hardware error/safety relay. - Wrong position Flexisoft/safety relay reported from autotruck system via CAN.	1) Restart the truck 2) Troubleshoot the CAN wiring 3) Reinstall the software 4) Replace the relay: Link target missing for title: [ID: 2700028043]
8:001	The collision sensor has registered a collision Collision sensor has registered collision levels outside set limit values as defined by parameters 105 and 106. Truck behaviour Creep speed	- Collision - Parameters 105 and 106 set too low.	1) Resetting truck after collision 2) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:002	The truck was intentionally taken out of service via I_site Truck behaviour Creep speed	•	1)
8:003	Argument error in option. Invalid configuration in one or more options. Truck behaviour Not influenced. The alarming option does not work.	Option argument error, parameters 1003 - 1042	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:004	Option buttons activated One or more option buttons on A5 is activated at start-up. Truck behaviour Not influenced. All option buttons are non-functional until the option button that is influenced is not influenced any more.	Option button pressed during start-up	1) Check your option buttons Check the function of the option buttons via the built-in test mode, Service Information mode 9 or TruckCom. If there is still a signal from a seemingly uninfluenced button, the button must be checked for jamming or unwanted connection.

No.	Description	Felorsak(er)	Action
8:010	CAN communication problem With the built-in charger T14 at start-up Truck behaviour Not influenced	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
8:011	CAN communication problem With the built-in charger T14. Truck behaviour Not influenced	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
8:020	Low voltage T14 detects no battery, voltage below valid level: 6V Truck behaviour Not influenced	Battery not connected to charger	1) Check the battery charger cables. Disconnect the battery by disconnecting the connector and power cable from the charger. Measure the ohms on red and black cable from T14 to X1. Replace the cables in case of breakage.
8:021	Low voltage T14 detects excessively low battery voltage when charging. Voltage below valid level: 19.8V Truck behaviour Not influenced	The battery is defective	1) Inspect the battery.
8:022	High voltage T14 detects excessively high battery voltage when charging. Voltage above valid level: 34.8V Truck behaviour Not influenced	The battery is defective	1) Inspect the battery.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
8:023	Low temperature T14 detects excessively low temperature when charging. Truck behaviour Not influenced	Truck used in cold environments.	1) Inform the client that the truck should be stored in normal ambient temperatures to avoid unnecessary wear and tear of the battery and other components.
8:024	High temperature T14 has reduced the charging current to 0A because of high temperature. Truck behaviour Not influenced	Cooling is reduced or the temperature sensor is defective.	1) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
8:027	Internal error in T14 Truck behaviour Not influenced	- No software compatibility - Defective component	1) Update the truck software Update the truck software to the latest version. 2) Replace the T14 See section "18.6 Built-in battery charger C5170".
8:028	Charging time The main charging time is exceeded by 5-11 hours depending of battery capacity. Truck behaviour Not influenced	- Battery parameters 109 and 110 incorrectly set. - The battery is defective	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 2) Inspect the battery.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
8:029	Charged capacity exceeds battery capacity. Truck behaviour Not influenced	- Battery parameters 109 and 110 incorrectly set. - The battery is defective	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly. 2) Inspect the battery.
8:030	Parameter 1101 not activated Communication with charger T14 exists, but factory parameter 1101 is not activated. Truck behaviour Not influenced	Incorrect value for parameter 1101	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:031	Saving parameter in T14 failed. Truck behaviour Not influenced	No software compatibility	1) Update the truck software Update the truck software to the latest version.
8:032	Wrong battery type selected via parameters 109 and 110. Truck behaviour Not influenced	Battery parameters 109 and 110 incorrectly set.	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:110	CAN communication problem between A5 and A36/A37 Truck behaviour Creep speed, blocked lifting/ lowering movement	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
8:111	CAN communication problem between A5 and A36/A37 Truck behaviour Creep speed, blocked lifting/ lowering movement	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
8:111	CAN communication problem between A5 and A36/A37 Truck behaviour Creep speed, blocked lifting/ lowering movement		chassis and CAN contact. Limit value >24 kOhm
8:112	CAN communication problem between A5 and B90 Truck behaviour Not influenced	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm
8:123	Argument error in option. Invalid configuration in one or more options. Truck behaviour Creep speed, blocked lifting/ lowering movement	Argument error option	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:130	CAN communication error CAN communications error with li-ion battery Truck behaviour Creep speed, blocked lifting movement If CAN key connected - creep speed	Error on CAN bus wiring or connections	1) Check the CAN bus Check that CAN is intact in terms of wiring harness, resistance and connectors. Disconnect the battery. Check that the resistance between (X41:3) and (X41:4) is 54 - 66 ohms. Check that the CAN wiring harness is not pinched. Measure resistance between the chassis and CAN contact. Limit value >24 kOhm 2) 3)
8:131	Temperature in the li-ion battery is too high Truck behaviour Creep speed, blocked lifting movement		1) Let the battery cool down

No.	Description	Felorsak(er)	Action
8:132	Current in the li-ion battery is too high Truck behaviour Creep speed, blocked lifting movement	Regenerative braking with fully-charged battery.	1) No action. 2) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50
8:133	Li-ion battery temperature below permitted level Truck behaviour Creep speed, blocked lifting movement		1) Move the truck to a warmer area
8:134	Defective hardware in the li-ion battery Truck behaviour Creep speed, blocked lifting movement	Defective sensor in the module.	1) Contact the battery manufacturer.
8:135	Internal CAN problem in the li-ion battery Truck behaviour Creep speed, blocked lifting movement	Communication problem between modules in the battery.	1) Contact the battery manufacturer. 2) 3)
8:136	Overvoltage in the li-ion battery Truck behaviour Creep speed, blocked lifting movement	A module in the battery has been switched off because its voltage is too high.	1) Restart the battery.
8:137	Battery cells deactivated by internal monitoring Truck behaviour Creep speed, blocked lifting movement	Some battery functions are not working.	1) Overvoltage, under-voltage, temperature too high, temperature too low. Let the battery rest and then restart the battery.
8:138	Low battery voltage in the li-ion battery Truck behaviour Creep speed, blocked lifting movement	- Battery discharged - Voltage drop.	1) Charge the battery 2) Check battery voltage Check the voltage using the built-in test procedure and comparing with value from external volt meter Check connectors, joints and splices for damage or corrosion. Check fuses F1 and F50

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
8:139	Wrong type of battery selected via parameter 1045 Truck behaviour Creep speed, blocked lifting movement	Battery parameter 1045 has incorrect setting.	1) Check parameters Check that factory parameters are set according to the truck configuration. Check that other parameters are set correctly.
8:140	Start-up problem between truck and li-ion battery during start-up. Truck behaviour Creep speed, blocked lifting movement	Internal program error	1) Update the truck software Create a truck report. Send a report to the manufacturer. Update the truck software to the latest version.
Cold	Low temperature The power stage of the motor control is very cold. The motor control temperature is below -20°C. The motor temperature sensor detects a low temperature. The motor temperature is below -35°C Truck behaviour The motor control reduces the maximum accelerating voltage linearly	- Truck used in cold environments - Cooling is reduced or the temperature sensor is defective.	1) Inform the client that the truck should be stored in normal ambient temperatures to avoid unnecessary wear and tear of the battery and other components. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.

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No.	Description	Felorsak(er)	Action
ESO	Emergency stop circuit open low voltage on T1:114 Truck behaviour Everything is stopped at once, the parking brake is activated, and the main contactor opens	- Emergency stop activated - Open electrical circuit	1) Return the emergency stop switch to the normal position and restart the truck 2) Check the circuit of the emergency stop Check that the emergency stop is in the normal position Check fuse F51 Check wiring and S21 after the break
Fold UP	The tiller arm comes down too quickly after start. Truck behaviour Driving stopped	The tiller arm comes down before the motor control unit is running.	1) Raise the arm.
Foot	Foot indicated outside of truck profile sensor B124/B125 Truck behaviour Driving stopped	- A foot is breaking the light barrier - Signal error	1) Keep the feet inside the truck profile. 2) Check the foot protection sensor Check whether the light source is misdirected Check the wiring harness for a break Check whether dirt or another object is blocking the light
gATE	Gate position The position of the gate in combination with the platform position and height sensor (SPE, SWE) is not permitted Truck behaviour Driving or lifting stopped	- The gate is not lowered when the platform is raised. - Gate in an undefined position - Gate raised when stacking above height limit 1.8 m	1) Lower the platform or lower the gate. 2) Move the gate to a defined position 3) Lower the gate for stacking above 1.8 meters in order to have three escape routes 4) Check the sensor signal Check that the distance between the sensor and the opposite part is correct Check that there is no dirt or foreign objects on the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct.

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Troubleshooting using error codes

No.	Description	Felorsak(er)	Action
Hot	High temperature The power stage in T1 shows a high temperature. The temperature is above 85°C. The motor temperature sensor detects a high temperature. The motor temperature exceeds +150°C Truck behaviour The motor control reduces the maximum accelerating voltage linearly	- Intensive use of the truck - Cooling is reduced or the temperature sensor is defective.	1) Intensive use of the truck. Let the truck cool down. 2) Check the cooling Check the cooling flanges of the transistor regulator, motor and steering motor as well as the battery charger and fans and filters for accumulations of dirt and dust. Too much dust and dirt cause the cooling not to work properly. Check that the fans work when the speed control is activated (fans behind T1 and motor fan) Check that the fan on charger T14 works when the charging begins, i.e. when the mains cable is connected Check the signal from the temperature sensors for T1 and T13 and from B1 using the built-in test or TruckCom. Determine if the temperature signals are plausible relative to the real temperatures of the components. If not, replace T1, T13 or replace B1 with a repair kit.
S-0h	Time for service Truck behaviour Not influenced	Time for service	1) After service is complete, enter a new or the same value in parameter 101
SLO	Reduced travel speed Truck behaviour Reduced speed and acceleration	Travel speed reduced because of activated Click-2-Creep, turtle or emergency drive mode.	1)
StOP	The platform is lowered but not loaded Truck behaviour Driving stopped	Platform in undefined position	1) Move the platform to a defined position 2) Check the sensor signal and/or adjust the platform Check that the distance between the sensor and the opposite part is correct Check that there is no dirt or foreign objects on

No.	Description	Felorsak(er)	Action
StOP	The platform is lowered but not loaded Truck behaviour Driving stopped		the sensor Inspect cabling. Measure the signals from the sensor with a volt meter and verify using TruckCom or the built-in test that the signal is correct.
tilt	Tilt angle The truck is tilting too much sideways when the forks reach the 1.8 meter sensor. Truck behaviour Lifting stopped	The truck is tilting sideways more than $\pm 1.5^\circ$ the same time the forks reach the 1.8 meter sensor.	Lower the forks below 1.8 meter.